

Problem 1 (Linear Regression) Let's implement a tool needed for a computer vision algorithm called photometric stereo, which allows us to reconstruct a 3D object's surface from 2D images of it under different lighting conditions. Figure 1 gives a sneak peek at what you'll be able to do after completing all the pieces. We want to measure the 3D surface reconstruction shown in Figure 1 right. We model the object's surface as $z = f(x, y) \in \mathbb{R}_+$ where $x, y \in \mathbb{R}$ denotes coordinates on the table, and z is the height of the surface above the table. To measure the object, we change the light source orientation $\mathbf{l} \in \mathbb{R}^{3 \times 1}$ (a unit-norm vector). For each orientation $\mathbf{l}$, we could obtain a pixel intensity image $I(x, y)$. Under a diffuse reflection assumption, the following relationship holds:

$$I(x, y) = \alpha(x, y) \mathbf{l}^T \mathbf{n}(x, y), \quad (1)$$

where $\alpha(x, y) \in \mathbb{R}$ is related to the surface's own property and $\mathbf{n}(x, y) \in \mathbb{R}^3$ is the unit-norm surface normal vector of $f(x, y)$. Assume that for each coordinate (x, y) in $f(x, y)$, it is corresponding to a pixel (x, y) in an $I(x, y)$. Also assume the light is directional light, i.e., the orientation $\mathbf{l}$ for the light is the same for all pixels (x, y) .

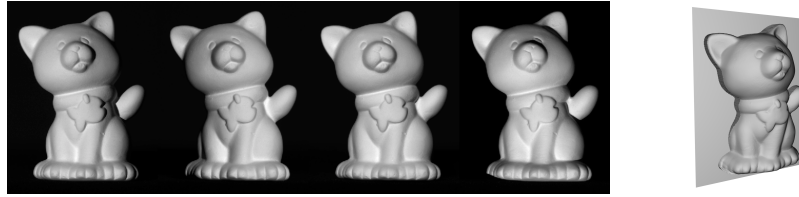


Figure 1: Photometric stereo. Left: 2D-pixel intensity $I(x, y)$ of a common scene under different lighting conditions. Right: 3D surface reconstruction $f(x, y)$ computed from the input images.

- (a) Suppose moving the light source in $\mathbf{l}_1, \dots, \mathbf{l}_d$, d different directions and obtain $\mathbf{I}_1, \dots, \mathbf{I}_d \in \mathbb{R}_+^{M \times N}$ 2D-pixel intensities matrix. We want to find the surface normal direction matrix $\mathbf{N} \in \mathbb{R}^{3 \times M \times N}$. Formulate this problem as a least squares problem.

- (b) Recovering the surface matrix $\mathbf{z} \in \mathbb{R}_+^{M \times N}$ from the surface normal direction matrix $\mathbf{N} \in \mathbb{R}^{3 \times M \times N}$. Formulate this problem as a least squares problem.

Hint 1: The relationship between surface norm $\mathbf{n}(x, y)$ and the object surface function $f(x, y)$ is:

$$\mathbf{n}(x, y) = \frac{\left[-\frac{\partial f(x, y)}{\partial x}, -\frac{\partial f(x, y)}{\partial y}, 1 \right]^T}{\sqrt{1 + \left(\frac{\partial f(x, y)}{\partial x} \right)^2 + \left(\frac{\partial f(x, y)}{\partial y} \right)^2}}.$$

Hint 2: The $f(x, y)$ could also be viewed as an image, such that:

$$\begin{aligned} \frac{\partial f(x, y)}{\partial x} &\approx \frac{f(x+1, y) - f(x, y)}{(x+1) - x} = f(x+1, y) - f(x, y) \\ \frac{\partial f(x, y)}{\partial y} &\approx \frac{f(x, y+1) - f(x, y)}{(y+1) - y} = f(x, y+1) - f(x, y) \end{aligned}$$